해답편

1) $\dfrac{\pi(\sqrt{3}+1)}{12}$

2)

가) $f'(0) = \lim_{h\to 0} \dfrac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \dfrac{\cancel{h}\tan^{-1}\left(\frac{1}{h}\right)}{\cancel{h}}$

$= \begin{cases} \lim_{h\to 0^+} \tan^{-1}\left(\frac{1}{h}\right) = \tan^{-1}(\infty) = \frac{\pi}{2} \\ \lim_{h\to 0^-} \tan^{-1}\left(\frac{1}{h}\right) = \tan^{-1}(-\infty) = -\frac{\pi}{2} \end{cases}$ $\quad \therefore \nexists f'(0)$

나) $f'(0) = \lim_{h\to 0} \dfrac{3\sin h + h^3\cos\left(\frac{1}{h^2}\right)}{h} = \lim_{h\to 0} \underbrace{\cancel{\dfrac{3\sin h}{h}}}_{1} + \boxed{h^2\cos\left(\frac{1}{h^2}\right)} = 3$

$|h^2\cos\frac{1}{h^2}| \le |h^2| \to 0$

squeeze에 의해 0으로 수렴!

다) $f'(0) = \lim_{h\to 0} \dfrac{(h^2-1)\sin|h|}{h}$

$= \begin{cases} \lim_{h\to 0^+} (h^2-1)\cdot \cancel{\dfrac{\sin h}{h}} = -1 \\ \lim_{h\to 0^-} \dfrac{(h^2-1)\cdot\sin(-h)}{h} = \lim_{h\to 0} -\cancel{\dfrac{(h^2-1)\sin h}{h}} = 1 \end{cases}$ $\quad \therefore \nexists f'(0)$

3) e

4) $-1+\dfrac{1}{e}$

5) $\dfrac{e}{2}$

6)

3. $\underbrace{\tan^{-1}2}_{\alpha} - \underbrace{\tan^{-1}\left(\frac{1}{3}\right)}_{\beta} \rightsquigarrow \tan(\alpha-\beta) = \dfrac{\tan\alpha - \tan\beta}{1+\tan\alpha\cdot\tan\beta}$

$\tan\alpha = 2 \quad \tan\beta = \frac{1}{3}$ $\quad = \dfrac{2-\frac{1}{3}}{1+2\cdot\frac{1}{3}}$

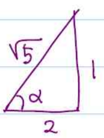
$= \dfrac{6-1}{3+2} = 1$

$\therefore \tan(\alpha+\beta) = 1 \qquad \alpha+\beta = \tan^{-1}(1) = \frac{\pi}{4}$

7)

$\underbrace{2\sin x + \cos x}$ $= \sqrt{5}\sin(x+\alpha)$

$\hookrightarrow \sqrt{5}\left(\underbrace{\dfrac{2}{\sqrt{5}}}_{\cos\alpha}\sin x + \underbrace{\dfrac{1}{\sqrt{5}}}_{\sin\alpha}\cdot\cos x\right)$



$\therefore \alpha = \cos^{-1}\left(\dfrac{2}{\sqrt{5}}\right)$ or $\alpha = \sin^{-1}\left(\dfrac{1}{\sqrt{5}}\right)$

8)

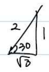
6. $\underbrace{\sin^{-1}\frac{1}{x}}_{\alpha} - \underbrace{\csc^{-1}x}_{\beta} = A$, $\quad \tan\left(\underbrace{\tan^{-1}\frac{1}{2}}_{\alpha} + \frac{\pi}{6}\right) = B$

$\sin\alpha = \frac{1}{x}$, $\csc\beta = x$

$\hookrightarrow \sin\beta = \frac{1}{x}$

$\therefore \alpha = \beta$, $\Rightarrow A = 0$

$\tan\alpha = \frac{1}{2}$



$\tan\left(\alpha+\frac{\pi}{6}\right) = \dfrac{\tan\alpha + \tan\frac{\pi}{6}}{1-\tan\alpha\cdot\tan\frac{\pi}{6}}$

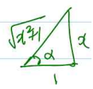
$= \dfrac{\frac{1}{2}+\frac{1}{\sqrt{3}}}{1-\frac{1}{2}\cdot\frac{1}{\sqrt{3}}}$

$= \dfrac{\frac{\sqrt{3}+2}{\cancel{2\sqrt{3}}}}{\frac{2\sqrt{3}-1}{\cancel{2\sqrt{3}}}} = \dfrac{\sqrt{3}+2}{2\sqrt{3}-1}$

$\therefore A+B = \dfrac{\sqrt{3}+2}{2\sqrt{3}-1}$

9)

7. $x+1 = 2\sin(2\underbrace{\tan^{-1}x}_{\alpha})$ $\quad \tan\alpha = x$



$\Rightarrow x+1 = 2\underbrace{\sin(2\alpha)}$ $\hookrightarrow 2\sin\alpha\cdot\cos\alpha$

$\rightarrow x+1 = 4\dfrac{x}{\sqrt{x^2+1}}\cdot\dfrac{1}{\sqrt{x^2+1}} = \dfrac{4x}{x^2+1}$

$\Rightarrow (x+1)(x^2+1) - 4x = 0$

$\rightarrow x^3 + x^2 - 3x + 1 = 0 \qquad \therefore -1$

$*\ (x-a)(x-b)(x-c) = 0$

$\hookrightarrow x^3 - (a+b+c)x^2 + (ab+bc+ca)x - abc = 0$

10)

$\ln\left\{4\underbrace{(\cosh^2 x - \sinh^2 x)}_{1}\right\} = \ln 4 = 2\ln 2$

11)

$\tan(x-y) = \dfrac{\tan x - \tan y}{1+\tan x\cdot\tan y}$

$\rightarrow -\tanh(x-y) = \dfrac{-\tanh x + \tanh y}{1 - \tanh x\cdot\tanh y}$

$\rightarrow \tanh(x+y) = \dfrac{\tanh x - \tanh y}{1-\tanh x\cdot\tanh y} = \dfrac{\frac{1}{2}-\left(-\frac{1}{2}\right)}{1-\frac{1}{2}\left(-\frac{1}{2}\right)} = \dfrac{1}{1+\frac{1}{4}}$

12)

$$f = x^{-1} \cdot \tan^{-1}(x^{-1})$$

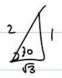
$$f' = -x^{-2} \cdot \tan^{-1}(x^{-1}) + x^{-1} \cdot \frac{1}{1+(x^{-1})^2} \cdot (-x^{-2})$$

$$f'(-1) = -\tan^{-1}(-1) + \frac{1}{2} = \frac{\pi}{4} + \frac{1}{2}$$

13)

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\cos t - (\cos t - t\sin t)}{\cos t - \sin t}$$

$$= \frac{t \cdot \sin t}{\cos t - \sin t}$$



$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{6}} = \frac{\frac{\pi}{6}\cdot \sin\frac{\pi}{6}}{\cos\frac{\pi}{6} - \sin\frac{\pi}{6}} = \frac{\frac{\pi}{6}\cdot\frac{1}{2}}{\frac{\sqrt{3}}{2} - \frac{1}{2}}$$

$$= \frac{\frac{\pi}{6}(\sqrt{3}+1)}{(\sqrt{3}-1)(\sqrt{3}+1)} = \frac{\frac{\pi}{6}(\sqrt{3}+1)}{2} = \frac{\pi}{12}(\sqrt{3}+1)$$

14)

5. $\ln f = x \cdot \ln(x^2+1)$

미분 $\Rightarrow \frac{1}{f} \cdot f' = \ln(x^2+1) + x \cdot \frac{2x}{x^2+1}$

$$f' = f\left[\ln(x^2+1) + \frac{2x^2}{x^2+1}\right]$$

$$f'(2) = f(2)\left[\ln 5 + \frac{8}{5}\right] = 40 + 5^2 \ln 5$$

15)

$h = f^{-1}$ $\quad f' = 1 + 3e^{3x}$

$$f^{-1\,\prime}(4) = \frac{1}{f'(f^{-1}(4))} = \frac{1}{1+3e^0} = \frac{1}{4}$$

$\underbrace{f^{-1}(4)}_{\alpha} \Leftrightarrow f(\alpha) = 4 \Leftrightarrow 3 + \alpha + e^{3\alpha} = 4$

$\therefore \alpha = 0$

16)

2. $e^a \sinh a = 2$ $\quad \mathrm{sech}\, 2a = \frac{1}{\cosh 2a}$

$\cos 2x = \cos^2 x - \sin^2 x$

$\cosh 2x = \cosh^2 x + \sinh^2 x$ $\quad = \frac{1}{\cosh^2 a + \sinh^2 a}$

$\sinh a = 2 \cdot e^{-a}$ $\quad -\sinh^2 a + \cosh^2 a = 1$

$\cosh^2 a = 1 + \sinh^2 a = 1 + 4e^{-2a}$

$$\mathrm{sech}\, 2a = \frac{1}{\cosh^2 a + \sinh^2 a} = \frac{1}{1+4e^{-2a}+4e^{-2a}} = \frac{1}{1+8e^{-2a}}$$

$$= \frac{e^{2a}}{e^{2a}+8} = \frac{5}{13}$$

$$\boxed{e^a \cdot \sinh a = 2 \Rightarrow e^a \cdot \frac{e^a - e^{-a}}{2} = 2 \Rightarrow e^{2a} - 1 = 4 \Rightarrow e^{2a} = 5}$$

17)

$$f' = \sinh^{-1}\left(\frac{x}{3}\right) + x \cdot \frac{1}{\sqrt{1+\left(\frac{x}{3}\right)^2}} \cdot \frac{1}{3} - \frac{2x}{2\sqrt{9+x^2}}$$

$$f'(3) = \underbrace{\sinh^{-1}(1)}_{\alpha} + \underbrace{\frac{1}{\sqrt{2}} - \frac{3}{3\sqrt{2}}}_{0} = \ln(1+\sqrt{2})$$

$\sinh \alpha = 1 \Rightarrow \frac{e^\alpha - e^{-\alpha}}{2} = 1 \Rightarrow e^{2\alpha} - 2e^\alpha - 1 = 0$

$\Rightarrow e^\alpha = 1 + \sqrt{2} \Rightarrow \alpha = \ln(1+\sqrt{2})$

18)

①
$$f'(0) = \lim_{h\to 0}\frac{f(0+h) - f(0)}{h} = \lim_{h\to 0}\frac{|h|\cdot\cosh - 0}{h}$$

$$= \begin{cases} \lim_{h\to 0^+}\frac{\cancel{h}\cdot\cosh}{\cancel{h}} = 1 \\ \lim_{h\to 0^-}\frac{-\cancel{h}\cosh}{\cancel{h}} = -1 \end{cases} \neq 1$$

$\therefore$ 존재 X

②
$$f'(0) = \lim_{h\to 0}\frac{|h|\sinh - 0}{h} = \begin{cases} \lim_{h\to 0^+}\frac{\cancel{h}\cdot\sinh}{\cancel{h}} = 0 \\ \lim_{h\to 0^-}\frac{-\cancel{h}\cdot\sinh}{\cancel{h}} = 0 \end{cases}$$

$\therefore$ 존재!

③
$$f'(0) = \lim_{h\to 0}\frac{|h|^3 - 0}{h} = \begin{cases} \lim_{h\to 0^+}\frac{h^3}{h} = 0 \\ \lim_{h\to 0^-}\frac{-h^3}{h} = 0 \end{cases}$$

$\therefore$ 존재!

④
$$f'(0) = \lim_{h\to 0}\frac{\cancel{h}|h| - 0}{\cancel{h}} = \begin{cases} \lim_{h\to 0^+} h = 0 \\ \lim_{h\to 0^-} -h = 0 \end{cases}$$

$\therefore$ 존재!

19)

1. ① $\cos 2x = \cos^2 x - \sin^2 x \Rightarrow \cosh 2x = \cosh^2 x + \sinh^2 x$

② $\dfrac{1+\tanh x}{1-\tanh x} = \dfrac{1+\frac{\sinh x}{\cosh x}}{1-\frac{\sinh x}{\cosh x}} = \dfrac{\cosh x + \sinh x}{\cosh x - \sinh x}$

$= \dfrac{\frac{e^x+e^{-x}}{2} + \frac{e^x-e^{-x}}{2}}{\frac{e^x+e^{-x}}{2} - \frac{e^x-e^{-x}}{2}} = \dfrac{2e^x}{2e^{-x}} = e^{2x}$ //

③ $\tanh(\ln x) = \dfrac{\sinh(\ln x)}{\cosh(\ln x)} = \dfrac{e^{\ln x}-e^{-\ln x}}{e^{\ln x}+e^{-\ln x}}$

$= \dfrac{x-\frac{1}{x}}{x+\frac{1}{x}} = \dfrac{x^2-1}{x^2+1}$ //

④ $\left[\tanh^{-1}(\sin x)\right]' = \dfrac{1}{1-\sin^2 x}\cdot(\sin x)' = \dfrac{\cos x}{1-\sin^2 x}$

20)

$\lim_{x\to 0} f(x) = \underset{a}{f(0)}$

$\lim_{x\to 0} \dfrac{\cos 3x - e^{4x}}{x} \overset{\frac{0}{0}\text{꼴!}}{=} \lim_{x\to 0}\left(-3\sin 3x - 4e^{4x}\right) = -4$

$\therefore a = -4$ //

21)

① $\lim_{x\to 1^+} (x-1)\tan\left(\frac{\pi x}{2}\right) = \lim_{x\to 1^+} \dfrac{\boxed{(x-1)}\cdot\boxed{\sin\left(\frac{\pi}{2}x\right)}}{\cos\left(\frac{\pi}{2}x\right)}$ $\to 0$, $\to 1$, $\to 0$

$= \lim_{x\to 1^+} \dfrac{x-1}{\cos\left(\frac{\pi}{2}x\right)} \overset{\left(\frac{0}{0}\text{꼴}\right)}{=} \lim_{x\to 1^+} \dfrac{1}{-\frac{\pi}{2}\sin\left(\frac{\pi}{2}x\right)} = -\dfrac{2}{\pi}$ //

② $\lim_{x\to 1} \dfrac{\sin(x-1)}{x^2+x-2} \overset{\left(\frac{0}{0}\right)}{=} \lim_{x\to 1} \dfrac{\cos(x-1)}{2x+1} = \dfrac{1}{3}$ //

③ $\lim_{x\to 0^+} \dfrac{\ln(\tan x)}{\ln(\sin x)} \overset{\left(\frac{\infty}{\infty}\right)}{=} \lim_{x\to 0^+} \dfrac{\frac{1}{\tan x}\cdot\sec^2 x}{\frac{1}{\sin x}\cdot\cos x} = \lim_{x\to 0^+} \dfrac{\frac{\cancel{\cos x}}{\cancel{\sin x}}\cdot\frac{1}{\cos^2 x}}{\frac{\cancel{\cos x}}{\cancel{\sin x}}}$

$= \lim_{x\to 0^+} \dfrac{1}{\cos^2 x} = 1$

④ $\lim_{x\to 0} \dfrac{\tan x}{x+\sin x} \overset{\left(\frac{0}{0}\right)}{=} \lim_{x\to 0} \dfrac{\sec^2 x}{1+\cos x} = \dfrac{1}{2}$

22)

$\lim_{x\to 0} \dfrac{e^x+e^{-x}-x^2-2}{\sin^2 x - x^2} \overset{\left(\frac{0}{0}\right)}{=} \lim_{x\to 0} \dfrac{e^x-e^{-x}-2x}{\underbrace{2\sin x\cdot\cos x}_{\sin 2x} - 2x}$

$\overset{\left(\frac{0}{0}\right)}{=} \lim_{x\to 0} \dfrac{e^x+e^{-x}-2}{2\cos 2x - 2} \overset{\left(\frac{0}{0}\right)}{=} \lim_{x\to 0} \dfrac{e^x-e^{-x}}{-4\sin 2x}$

$\overset{\left(\frac{0}{0}\right)}{=} \lim_{x\to 0} \dfrac{e^x+e^{-x}}{-8\cos 2x} = \dfrac{2}{-8} = -\dfrac{1}{4}$ //

23)

$\lim_{x\to 0^+} e^{\ln(\tan^{-1}x)^{\frac{1}{\ln x}}} = e^{\boxed{\lim_{x\to 0^+} \frac{1}{\ln x}\cdot \ln(\tan^{-1}x)}}$

$\lim_{x\to 0^+} \dfrac{\ln(\tan^{-1}x)}{\ln x} \overset{\left(\frac{\infty}{\infty}\right)}{=} \lim_{x\to 0^+} \dfrac{\frac{1}{\tan^{-1}x}\cdot\frac{1}{1+x^2}}{\frac{1}{x}}$

$= \lim_{x\to 0^+} \dfrac{x}{\boxed{(1+x^2)}\tan^{-1}x} = \lim_{x\to 0^+} \dfrac{x}{\tan^{-1}x}$ $\to 1$

$\overset{\left(\frac{0}{0}\right)}{=} \lim_{x\to 0^+} \dfrac{1}{\frac{1}{1+x^2}} = \lim_{x\to 0^+}(1+x^2) = 1$

$\therefore e^1$ //

24)

① $f = x - \tan^{-1}x \qquad f' = 1 - \dfrac{1}{1+x^2} = \dfrac{x^2}{1+x^2} > 0 \Rightarrow f: \uparrow$

$f(0) = 0$

$\therefore f(x) > 0$ 즉, $x > \tan^{-1}x$ //

② $f = \tan^{-1}x - \dfrac{x}{1+x^2} \qquad f' = \dfrac{1}{1+x^2} - \dfrac{(1+x^2) - x\cdot(2x)}{(1+x^2)^2}$

$= \dfrac{2x^2}{(1+x^2)^2} > 0 \Rightarrow f: \uparrow$

$f(0) = 0 \qquad \therefore f(x) > 0$ 즉, $\tan^{-1}x > \dfrac{x}{1+x^2}$

③ $f = x - \sin^{-1}x \qquad f' = 1 - \dfrac{1}{\sqrt{1-x^2}} = \dfrac{\sqrt{1-x^2}-1}{\sqrt{1-x^2}} < 0 \quad (\because 0<x<1)$

$f(0) = 0 \qquad \therefore f(x) < 0$ 즉, $\sin^{-1}x > x$ //

25)

$f' = \left(\displaystyle\int_{\sqrt{x}}^{x} \dfrac{e^t}{t^2}\,dt\right)' = \dfrac{e^x}{x^2} - \dfrac{e^{\sqrt{x}}}{x}(\sqrt{x})'$

$= \dfrac{e^x}{x^2} - \dfrac{e^{\sqrt{x}}}{x}\cdot\dfrac{1}{2\sqrt{x}}$

$f'(1) = e - e\cdot\dfrac{1}{2} = \dfrac{e}{2}$ //

26)

$$f' = \left(\int_{\tan x}^{1} \frac{1}{\sqrt{t^2+3}}\,dt\right)' = -\frac{1}{\sqrt{\tan^2 x+3}}\cdot(\tan x)'$$

$$= -\frac{1}{\sqrt{\tan^2 x+3}}\sec^2 x$$

$$f'\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{\tan^2\frac{\pi}{4}+3}}\cdot\frac{1}{\cos^2\frac{\pi}{4}} = -\frac{1}{2}\cdot 2 = -1$$

27)

$$f(x) = \int_1^x \left(e^{-x}\ln t - t\ln t\right)dt$$

$$= e^{-x}\int_1^x \ln t\,dt - \int_1^x t\ln t\,dt$$

$$f' = -e^{-x}\int_1^x \ln t\,dt + e^{-x}\ln x - x\ln x$$

$$f'' = e^{-x}\int_1^x \ln t\,dt - e^{-x}\ln x - e^{-x}\ln x + e^{-x}\frac{1}{x} - \ln x - x\cdot\frac{1}{x}$$

$$f''(1) = e^{-1}\underbrace{\int_1^1 \ln t\,dt}_{=0} - \underbrace{e^{-1}\ln 1 - e^{-1}\ln 1}_{=0} + e^{-1} - \underbrace{\ln 1}_{=0} - 1 = -1+\frac{1}{e}$$

28)

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{\cancel{f(x)}+f(h)+2xh-\cancel{f(x)}}{h}$$

$$= \lim_{h\to 0}\left(\frac{f(h)}{h}+2x\right) = 5+2x$$

29)

3, $f(x) = \ln(3x+1)^{-\frac{1}{3}} = -\frac{1}{3}\ln(3x+1)$

$$f' = -\frac{1}{\cancel{3}}\cdot\frac{\cancel{3}}{3x+1} = -(3x+1)^{-1}$$

$$f'' = (3x+1)^{-2}\cdot 3 \qquad f''' = -3^2\cdot 2(3x+1)^{-3}$$

$$f^{(4)} = 3^3\cdot\overbrace{2\cdot 3}^{3!}(3x+1)^{-4}$$

$$\cdots\ f^{(n)} = (-1)^n\,3^{(n-1)}(n-1)!\,(3x+1)^{-n}$$

$$\therefore f^{(100)} = 3^{99}\cdot 99!\,(3x+1)^{-100}$$

30)

$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e$ 을 이용하자.

$\frac{a}{n} \triangleq \frac{1}{N}$ 라 두자. $N = \frac{n}{a} \Rightarrow n = aN$

$$\lim_{N\to\infty}\left(1+\frac{1}{N}\right)^{2\cdot aN} = \left[\lim_{N\to\infty}\left(1+\frac{1}{N}\right)^N\right]^{2a} = e^{2a}$$

∴ 3번

31)

실수 x에 대하여 $[x]\le x<[x]+1$이므로

$0\le x-[x]<1$

$\therefore 3\le x-[x]+3<4$ 그런데 $\left[\frac{x+3}{2}\right]$이 정수이므로 $\left[\frac{x+3}{2}\right]=3$

$\therefore 3\le\frac{x+3}{2}<4$

$\Rightarrow 6\le x+3<8 \Rightarrow 3\le x<5$

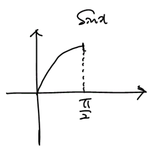
따라서 $[x]$의 값은 3 또는 4이다.

$\therefore 3+4=7$

정답 ③

32)

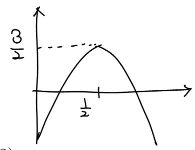
$$y = 2\sin x + 1 - 2\sin^2 x$$



$\sin x \triangleq t \Rightarrow 0\le t\le 1$

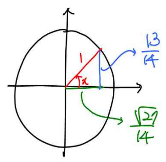
$$y = -2t^2+2t+1 = -2(t^2-t)+1$$

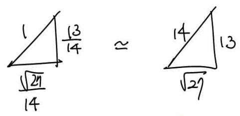
$$= -2\left(t-\frac{1}{2}\right)^2+\frac{3}{2}$$



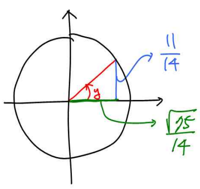
∴ 최대 $= \frac{3}{2}$ 5번

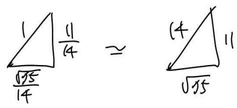
33)





$\therefore \tan x = \frac{13}{\sqrt{27}} = \frac{13}{3\sqrt{3}}$

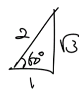




$\therefore \tan y = \frac{11}{\sqrt{75}} = \frac{11}{5\sqrt{3}}$

$$\tan(x+y) = \frac{\tan x+\tan y}{1-\tan x\tan y} = \frac{\frac{13}{3\sqrt{3}}+\frac{11}{5\sqrt{3}}}{1-\frac{13}{3\sqrt{3}}\cdot\frac{11}{5\sqrt{3}}} = \frac{13\times5\sqrt{3}+11\times3\sqrt{3}}{15\times3-13\times11} = -\sqrt{3}$$

34)



$$\lim_{x\to\frac{\pi}{3}} f(x) = \begin{cases}\lim_{x\to\frac{\pi}{3}+} x+a = \frac{\pi}{3}+a \\ \lim_{x\to\frac{\pi}{3}-} a\sec x = a\cdot\frac{1}{\boxed{\cos\frac{\pi}{3}}} = 2a\end{cases}$$

$\frac{\pi}{3}+a = 2a \Rightarrow a = \frac{\pi}{3}$ ∴ 2번

35)

$$f'(0) = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h} = \lim_{h\to 0}\frac{2\cancel{h}|h| - 0}{\cancel{h}}$$

$$= \lim_{h\to 0} 2|h| = \begin{cases} \lim_{h\to 0^+}(2h) = 0 \\ \lim_{h\to 0^-}(-2h) = 0 \end{cases} \quad \therefore f'(0)=0$$

2번

36)

① $f'(0) = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h} = \lim_{h\to 0}\frac{|h|\cdot\sin h - 0}{h}$

$$= \begin{cases} \lim_{h\to 0^+}\frac{\cancel{h}\sin h}{\cancel{h}} = 0 \\ \lim_{h\to 0^-}\frac{-\cancel{h}\cdot\sin h}{\cancel{h}} = 0 \end{cases} \quad \therefore \text{미분가능}$$

2번

② $f'(0) = \lim_{h\to 0}\frac{|h|\cdot\cosh - 0}{h} = \begin{cases} \lim_{h\to 0^+}\frac{\cancel{h}\cdot\cosh}{\cancel{h}} = 1 \\ \lim_{h\to 0^-}\frac{-\cancel{h}\cdot\cosh}{\cancel{h}} = -1 \end{cases}$ $\therefore$ 미분불가능

37) 함수 $f(x)$가 $x=0$에서 연속이려면 $\lim_{x\to 0}\frac{\sqrt{x^2+4}+a}{x^2} = b$

위의 식에서 $\lim_{x\to 0}x^2 = 0$이므로 $\lim_{x\to 0}(\sqrt{x^2+4}+a) = 2+a = 0$

$\therefore\ a = -2$

이를 주어진 식에 대입하면

$$\lim_{x\to 0}\frac{\sqrt{x^2+4}-2}{x^2} = \lim_{x\to 0}\left(\frac{\sqrt{x^2+4}-2}{x^2}\times\frac{\sqrt{x^2+4}+2}{\sqrt{x^2+4}+2}\right)$$

$$= \frac{1}{4} = b$$

$\therefore b = \frac{1}{4}$ $\quad \therefore\ ab = -\frac{1}{2}$

정답 ②

38) (i) $0 \le x < 1$일 때 $f(x) = -ax^2+2$

(ii) $x > 1$일 때 $f(x) = x$

(iii) $x = 1$일 때 $f(1) = \frac{3-a}{2}$

함수 $f(x)$가 $x=1$에서 연속이려면

$\lim_{x\to 1-}(-ax^2+2) = \lim_{x\to 1+}x = f(1)$

$-a+2 = 1 = \frac{3-a}{2}$ $\quad \therefore\ a = 1$

정답 ③

39) $f'(x) = \frac{g(x)+3-xg'(x)}{\{g(x)+3\}^2}$

$f'(0) = 1$에서 $\frac{g(0)+3}{\{g(0)+3\}^2} = 1$, $\frac{1}{g(0)+3} = 1$, $g(0)+3 = 1$

$\therefore g(0) = -2$

정답 ①

40) (준식)의 양변에 자연로그를 취하면

$f(x) = \frac{1}{2}\ln\frac{1-\cos x}{1+\cos x} = \frac{1}{2}\{\ln(1-\cos x) - \ln(1+\cos x)\}$

이제 양변을 x에 대하여 미분하면

$f'(x) = \frac{1}{2}\left(\frac{\sin x}{1-\cos x} - \frac{-\sin x}{1+\cos x}\right) = \frac{2\sin x}{2(1-\cos^2 x)}$

$= \frac{1}{\sin x} = \csc x$

정답 ④

41) $g'(x) = -\frac{(1-xf(x))'}{(1-xf(x))^2} = \frac{f(x)+xf'(x)}{(1-xf(x))^2}$

$\therefore g'(0) = \frac{f(0)+xf'(0)}{(1-0\cdot f(0))^2} = f(0) = 1$

정답 ③

42) $f(0) = \sin^{-1}(0) = 0$이므로

$\lim_{h\to 0}\frac{f(0+2h)-f(0)}{h} = 2\lim_{h\to 0}\frac{f(0+2h)-f(0)}{2h}$

$= 2f'(0)$

$f'(x) = \frac{1}{\sqrt{1-x^2}}$ 이므로 $f'(0) = 1$

$\therefore\ 2f'(0) = 2$

정답 ①

43) $(f\circ f)(x) = f(f(x)) = f(ax+b) = a(ax+b)+b = a^2x+ab+b$

따라서 $a^2 = 4,\ ab+b = 6$이므로 $a = 2,\ b = 2(\because\ a > 0)$

$\therefore\ f(x) = 2x+2$

정답 ①

44) $\lim_{x\to 3^-}\frac{2x}{x-3} = -\infty$ 이고 $\lim_{x\to 3^+}\frac{2x}{x-3} = +\infty$

정답 ①

45) $f(x) = \frac{x^{\frac{3}{4}}\sqrt{x^2+1}}{(3x+2)^5}$

양변에 로그를 취하면 $\ln f(x) = \ln\frac{x^{\frac{3}{4}}\sqrt{x^2+1}}{(3x+2)^5}$

$= \frac{3}{4}\ln x + \frac{1}{2}\ln(x^2+1) - 5\ln(3x+2)$

양변을 미분하면 $\frac{f'(x)}{f(x)} = \frac{3}{4x} + \frac{2x}{2(x^2+1)} - \frac{15}{3x+2}$

따라서 $f'(x) = \frac{x^{\frac{3}{4}}\sqrt{x^2+1}}{(3x+2)^5}\left(\frac{3}{4x} + \frac{x}{x^2+1} - \frac{15}{3x+2}\right)$

정답 ③

46) $\lim_{x\to\infty}(\sqrt{x^2+x} - x) = \lim_{x\to\infty}\frac{(\sqrt{x^2+x} - x)(\sqrt{x^2+x} + x)}{(\sqrt{x^2+x} + x)}$

$= \lim_{x\to\infty}\frac{x}{(\sqrt{x^2+x} + x)} = \frac{1}{2}$

정답 ③

47) $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{1-\frac{2}{t}}{e^{\sqrt{t}}\frac{1}{2\sqrt{t}}}$

$t = 1$을 대입하면 $\frac{dy}{dx} = -\frac{2}{e}$

정답 ④